Appendix I: Handouts-2

Row Echelon Form (REF) and Reduced Row Echelon Form (RREF) and Rank of a Matrix, Solution of System of Equations using Gaussian and Gauss-Jordan Elimination Method and Homogeneous System of Equations

---

1. Row Echelon Form (REF)

Definition

A matrix is said to be in **Row Echelon Form (REF)** if:

1. All zero rows (if any) are at the bottom of the matrix.
2. The leading entry (the first non-zero entry from the left) of each non-zero row is to the right of the leading entry of the row above it.
3. All entries below each leading entry are zero.

Important Remark

The leading entries **need not be 1**. They may be any non-zero numbers.

Example 1 (REF)

$$\begin{pmatrix} 2 & 3 & 1 \\ 0 & 5 & 4 \\ 0 & 0 & 7 \end{pmatrix}$$

Leading entries are (2,5,7).

The pivots move to the right and all entries below them are zero.

Hence it is in REF.

Example 2 (REF)

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 4 & 5 & 6 \\ 0 & 0 & 8 & 9 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

This is also in REF.

Non-Example 1

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 0 & 1 & 2 \end{pmatrix}$$

The entry below the first leading entry is not zero.

Therefore it is not in REF.

Non-Example 2

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 4 \end{pmatrix}$$

A zero row appears above a non-zero row.

Therefore it is not in REF.

---

2. Reduced Row Echelon Form (RREF)

Definition

A matrix is said to be in **Reduced Row Echelon Form (RREF)** if:

1. It is already in REF.
2. Every leading entry is 1.
3. Each leading 1 is the only non-zero entry in its column.

In other words:

- All entries below a pivot are zero.
- All entries above a pivot are zero.

Example 1 (RREF)

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

Each pivot is 1 and is the only non-zero entry in its column.

Hence it is in RREF.

Example 2 (RREF)

$$\begin{pmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 6 \end{pmatrix}$$

This is also in RREF.

Non-Example 1

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{pmatrix}$$

This matrix is in REF but not in RREF because there are non-zero entries above pivots.

Non-Example 2

$$\begin{pmatrix} 2 & 0 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{pmatrix}$$

The first pivot is 2, not 1.

Therefore it is not in RREF.

Comparison Between REF and RREF

Property	REF	RREF
Zero rows at bottom	✓	✓
Leading entries move right	✓	✓
Entries below pivots are zero	✓	✓
Leading entries must be 1	No	Yes
Entries above pivots are zero	Not required	Required
Unique form	No	Yes
Easier to obtain	Yes	No
Used in Gaussian Elimination	Yes	Yes
Used in Gauss–Jordan Elimination	No	Yes

Important Relationship

$$\boxed{\text{Every RREF matrix is a REF matrix.}}$$

However,

$$\boxed{\text{Not every REF matrix is an RREF matrix.}}$$

Student Practice Section

For each of the following matrices, determine whether it is:

1. Not in Row Echelon Form (REF),
2. In Row Echelon Form (REF) but not in Reduced Row Echelon Form (RREF), or
3. In Reduced Row Echelon Form (RREF).

Provide a justification for your classification.

(a)

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{pmatrix}$$

(b)

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

(c)

$$\begin{pmatrix} 2 & 3 & 1 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$$

(d)

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 4 & 5 \\ 0 & 1 & 2 \end{pmatrix}$$

(e)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(f)

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 3 \end{pmatrix}$$

(g)

$$\begin{pmatrix} 1 & 0 & 4 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

(h)

$$\begin{pmatrix} 3 & 2 & 1 \\ 0 & 5 & 4 \\ 0 & 0 & 2 \end{pmatrix}$$

(i)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}$$

(j)

$$\begin{pmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Quick Memory Trick

REF = Staircase Form

Think of the pivots as forming steps:

REF

```
Step 1: Build the staircase

■ * * *
0 ■ * *
0 0 ■ *
0 0 0 ■
```

RREF = Clean Staircase Form

The staircase is completely cleaned:

RREF

```
Step 2: Clean the staircase

1 0 0 *
0 1 0 *
0 0 1 *
0 0 0 0
```

Every pivot column contains exactly one non-zero entry, namely the pivot 1.

$$\boxed{\text{REF: Zeros below pivots}}$$

$$\boxed{\text{RREF: Zeros above and below pivots}}$$

Example 1

Convert

$$A = \begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix}$$

to REF and then to RREF.

Step 1: Convert to REF

Apply

$$R_2 \to R_2 - \frac{1}{2}R_1$$

$$\begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix} \to \begin{pmatrix} 2 & 4 \\ 0 & 1 \end{pmatrix}$$

This is in **REF**.

Step 2: Convert REF to RREF

The pivot in Row 2 is already 1.

Now eliminate the entry above it:

$$R_1 \to R_1 - 4R_2$$

$$\begin{pmatrix} 2 & 4 \\ 0 & 1 \end{pmatrix} \to \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

Now make the first pivot equal to 1:

$$R_1 \to \frac{1}{2}R_1$$

$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Hence the **RREF** is

$$\boxed{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}$$

Example 2

Convert

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}$$

to REF and then to RREF.

Step 1: Convert to REF

Apply

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - 3R_1$$

giving

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

This is in **REF**.

Step 2: Convert REF to RREF

The pivot in Row 2 is already 1.

Eliminate the entry above it:

$$R_1 \to R_1 - 2R_2$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

Now every pivot column contains a single non-zero entry.

Hence the **RREF** is

$$\boxed{\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}}$$

Observation for Students

Example 1

REF:

$$\begin{pmatrix} 2 & 4 \\ 0 & 1 \end{pmatrix}$$

RREF:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Example 2

REF:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

RREF:

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

Example 3 (4 × 4 Matrix)

Convert the matrix

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 5 & 3 & 1 \\ 3 & 8 & 5 & 2 \\ 4 & 11 & 7 & 3 \end{pmatrix}$$

to REF and RREF.

Step 1: Eliminate entries below the first pivot

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - 3R_1$$

$$R_4 \to R_4 - 4R_1$$

gives

$$\begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 3 & 3 & 3 \end{pmatrix}$$

Step 2: Eliminate below the second pivot

$$R_3 \to R_3 - 2R_2$$

$$R_4 \to R_4 - 3R_2$$

$$\begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

This is the **REF**.

$$\boxed{REF = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}$$

Step 3: Convert REF to RREF

Eliminate the 2 above the second pivot:

$$R_1 \to R_1 - 2R_2$$

$$\begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Thus

$$\boxed{RREF = \begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}$$

Example 4 (4 × 4 Matrix with Full Rank)

Convert

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

to REF and RREF.

Notice that this matrix is already in REF.

$$\boxed{REF = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}}$$

Convert to RREF

$$R_3 \to R_3 - R_4$$

$$R_2 \to R_2 - R_3$$

$$R_1 \to R_1 - R_2$$

After simplification,

$$\boxed{RREF = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}$$

Example 5 (2 × 3 Matrix)

Convert

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \end{pmatrix}$$

to REF and RREF.

REF

$$R_2 \to R_2 - 2R_1$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix}$$

Thus

$$\boxed{REF = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix}}$$

RREF

$$R_1 \to R_1 - 2R_2$$

$$\boxed{RREF = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}}$$

Example 6 (3 × 4 Matrix)

Convert

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 4 & 5 \\ 3 & 5 & 7 & 9 \end{pmatrix}$$

to REF and RREF.

REF

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - 3R_1$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 2 & 4 & 6 \end{pmatrix}$$

Now

$$R_3 \to R_3 - 2R_2$$

$$\boxed{REF = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}}$$

RREF

$$R_1 \to R_1 - R_2$$

$$\boxed{RREF = \begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}}$$

Example 7 (4 × 5 Matrix)

Convert

$$A = \begin{pmatrix} 1 & 1 & 0 & 2 & 1 \\ 2 & 3 & 1 & 5 & 3 \\ 3 & 4 & 1 & 7 & 4 \\ 4 & 5 & 1 & 9 & 5 \end{pmatrix}$$

to REF and RREF.

Step 1: Eliminate entries below the first pivot

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - 3R_1$$

$$R_4 \to R_4 - 4R_1$$

giving

$$\begin{pmatrix} 1 & 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \end{pmatrix}$$

Step 2: Eliminate below the second pivot

$$R_3 \to R_3 - R_2$$

$$R_4 \to R_4 - R_2$$

Hence

$$\boxed{REF = \begin{pmatrix} 1 & 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}}$$

Step 3: Convert REF to RREF

$$R_1 \to R_1 - R_2$$

Thus

$$\boxed{RREF = \begin{pmatrix} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}}$$

Example 8 (5 × 4 Matrix)

Convert

$$A = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 2 & 1 & 3 & 5 \\ 3 & 1 & 4 & 7 \\ 4 & 2 & 6 & 10 \\ 5 & 2 & 7 & 12 \end{pmatrix}$$

to REF and RREF.

Step 1

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - 3R_1$$

$$R_4 \to R_4 - 4R_1$$

$$R_5 \to R_5 - 5R_1$$

giving

$$\begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 2 & 2 & 2 \end{pmatrix}$$

Step 2

$$R_3 \to R_3 - R_2$$

$$R_4 \to R_4 - 2R_2$$

$$R_5 \to R_5 - 2R_2$$

Therefore

$$\boxed{REF = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}$$

Step 3

There are already zeros above the second pivot.

Therefore

$$\boxed{RREF = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}$$

Example 9 (5 × 5 Matrix)

Convert

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

to REF and RREF.

Notice that the matrix is already in REF.

$$\boxed{REF = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}}$$

Convert to RREF

$$R_4 \to R_4 - R_5$$

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$R_3 \to R_3 - R_4$$

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$R_2 \to R_2 - R_3$$

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$R_1 \to R_1 - R_2$$

Hence

$$\boxed{RREF = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}}$$

Exercises

Convert each matrix to REF and RREF.

Exercise 1

$$\begin{pmatrix} 1 & 2 & 1 & 0 & 3 \\ 2 & 5 & 3 & 1 & 8 \\ 3 & 8 & 5 & 2 & 13 \\ 4 & 11 & 7 & 3 & 18 \end{pmatrix}$$

Exercise 2

$$\begin{pmatrix} 1 & 0 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 8 \\ 3 & 1 & 7 & 4 & 11 \\ 4 & 2 & 10 & 6 & 16 \end{pmatrix}$$

Exercise 3

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 3 \\ 3 & 8 & 2 & 5 \\ 4 & 11 & 3 & 7 \\ 5 & 14 & 4 & 9 \end{pmatrix}$$

Exercise 4

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 4 & 5 \\ 3 & 5 & 7 & 9 \\ 4 & 7 & 10 & 13 \\ 5 & 9 & 13 & 17 \end{pmatrix}$$

Exercise 5

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 2 & 3 & 1 & 0 & 0 \\ 3 & 5 & 3 & 1 & 0 \\ 4 & 7 & 6 & 3 & 1 \\ 5 & 9 & 10 & 6 & 3 \end{pmatrix}$$

Rank of a Matrix

Definition

The **rank** of a matrix is the maximum number of linearly independent rows or columns of the matrix.

Equivalently,

The rank of a matrix is the number of **non-zero rows in its Row Echelon Form (REF)** or Reduced Row Echelon Form (RREF).

Notation

If A is a matrix, its rank is denoted by

rank(A)

or simply

r(A).

Why is Rank Important?

The rank tells us:

- How much independent information is contained in a matrix.
- Whether the rows or columns are linearly dependent.
- Whether a system of linear equations has a solution.
- Whether a square matrix is invertible.

Example 1: Full Rank Matrix

Consider

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Convert to REF:

$$R_2 \to R_2 - 3R_1$$

$$\begin{pmatrix} 1 & 2 \\ 0 & -2 \end{pmatrix}$$

There are **2 non-zero rows.**

Therefore,

$$\boxed{\text{rank}(A) = 2}$$

Since the matrix is 2×2, the rank equals the order of the matrix.

Example 2: Rank 1 Matrix

Consider

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

Notice that

$$R_2 = 2R_1.$$

Apply

$$R_2 \to R_2 - 2R_1$$

$$\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$$

There is only **one non-zero row.**

Hence,

$$\boxed{\text{rank}(A) = 1}$$

Example 3: A 3×3 Matrix

Consider

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{pmatrix}$$

Apply

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - R_1$$

giving

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{pmatrix}$$

Interchange R_2 and R_3:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$$

There are **2 non-zero rows.**

Therefore,

$$\boxed{\text{rank}(A) = 2}$$

Example 4: Identity Matrix

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

There are 3 pivots and 3 non-zero rows.

Thus

$$\boxed{\text{rank}(I_3) = 3}$$

Example 5: Zero Matrix

$$O = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

There are no non-zero rows.

Hence

$$\boxed{\operatorname{rank}(O) = 0}$$

Useful Facts About Rank

Fact 1

For an $m \times n$ matrix,

$$\boxed{\operatorname{rank}(A) \le \min(m, n)}$$

Fact 2

A matrix has full rank if

$$\operatorname{rank}(A) = \min(m, n).$$

Examples:

- 3×3 matrix → full rank = 3
- 4×5 matrix → full rank = 4
- 5×4 matrix → full rank = 4

Fact 3

For a square matrix A,

$$A \text{ is invertible} \iff \operatorname{rank}(A) = n.$$

Fact 4

If

$$\det(A) \neq 0,$$

then

$$\boxed{\operatorname{rank}(A) = n}$$

for an $n \times n$ matrix.

Exercise Set: Rank of a Matrix

Instructions

For each matrix:

1. Convert the matrix to Row Echelon Form (REF).
2. Convert the matrix to Reduced Row Echelon Form (RREF).
3. Hence determine the rank of the matrix.

1.

$$A = \begin{pmatrix} 1 & 1 & 0 & 2 & 1 \\ 2 & 3 & 1 & 5 & 3 \\ 3 & 4 & 1 & 7 & 4 \\ 4 & 5 & 1 & 9 & 5 \end{pmatrix}$$

2.

$$A = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 2 & 1 & 3 & 5 \\ 3 & 1 & 4 & 7 \\ 4 & 2 & 6 & 10 \\ 5 & 2 & 7 & 12 \end{pmatrix}$$

3.

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

4.

$$\begin{pmatrix} 1 & 2 & 1 & 0 & 3 \\ 2 & 5 & 3 & 1 & 8 \\ 3 & 8 & 5 & 2 & 13 \\ 4 & 11 & 7 & 3 & 18 \end{pmatrix}$$

5.

$$\begin{pmatrix} 1 & 0 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 8 \\ 3 & 1 & 7 & 4 & 11 \\ 4 & 2 & 10 & 6 & 16 \end{pmatrix}$$

6.

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 3 \\ 3 & 8 & 2 & 5 \\ 4 & 11 & 3 & 7 \\ 5 & 14 & 4 & 9 \end{pmatrix}$$

7.

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 4 & 5 \\ 3 & 5 & 7 & 9 \\ 4 & 7 & 10 & 13 \\ 5 & 9 & 13 & 17 \end{pmatrix}$$

8.

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 2 & 3 & 1 & 0 & 0 \\ 3 & 5 & 3 & 1 & 0 \\ 4 & 7 & 6 & 3 & 1 \\ 5 & 9 & 10 & 6 & 3 \end{pmatrix}$$

9.

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 6 & 8 & 10 \\ 3 & 6 & 9 & 12 & 15 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

10.

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 5 & 5 & 5 & 5 \end{pmatrix}$$

Solve the following system of linear equations using

i) **Gaussian Elimination Method** and

ii) **Gauss-Jordan Elimination Method**

Problem-1:

$x + y = 3$

$x - y = 1$

Solution:

Augmented Matrix

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 1 & -1 & 1 \end{array}\right]$$

Method 1: Gaussian Elimination

Step 1

$$R_2 \to R_2 - R_1$$

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -2 & -2 \end{array}\right]$$

This is REF.

Step 2: Back Substitution

From row 2,

$$-2y = -2$$

$$y = 1$$

Substituting into row 1,

$$x + 1 = 3$$

$$x = 2$$

Solution

$$\boxed{x = 2, \qquad y = 1}$$

Method 2: Gauss-Jordan Elimination

Starting from

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 1 & -1 & 1 \end{array}\right]$$

Step 1

$$R_2 \to R_2 - R_1$$

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -2 & -2 \end{array}\right]$$

Step 2

$$R_2 \rightarrow -\frac{1}{2}R_2$$

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 1 \end{array}\right]$$

Step 3

$$R_1 \rightarrow R_1 - R_2$$

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right]$$

This is RREF.

Solution

$$\boxed{x = 2, \qquad y = 1}$$

Problem-2:

$x + y = 2$

$2x + 2y = 4$

Solution:

Augmented Matrix

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & 2 & 4 \end{array}\right]$$

Method 1: Gaussian Elimination

Step 1

$$R_2 \to R_2 - 2R_1$$

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 0 \end{array}\right]$$

This is REF.

Interpretation

The second row gives

$$0 = 0.$$

This equation provides no new information.

Thus

$$x + y = 2.$$

Let

$$y = t$$

where t is any real number.

Then

$$x = 2 - t.$$

Solution Set

$$\boxed{x = 2 - t, \qquad y = t}$$

where $t \in \mathbb{R}$.

Method 2: Gauss-Jordan Elimination

The matrix

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 0 \end{array}\right]$$

is already in RREF.

Therefore

$$x + y = 2.$$

Let

$$y = t.$$

Then

$$x = 2 - t.$$

Solution Set

$$\boxed{(x, y) = (2 - t, t), \quad t \in \mathbb{R}}$$

Problem-3:

$x + y = 2$

$x + y = 3$

Solution:

Augmented Matrix

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 1 & 1 & 3 \end{array}\right]$$

Method 1: Gaussian Elimination

Step 1

$$R_2 \to R_2 - R_1$$

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 1 \end{array}\right]$$

This is REF.

Interpretation

The second row means

$$0x + 0y = 1$$

or

$$0 = 1.$$

This is impossible.

Conclusion

$$\boxed{\text{No Solution}}$$

The system is **inconsistent**.

Method 2: Gauss-Jordan Elimination

The matrix

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 1 \end{array}\right]$$

is already in RREF.

The second row represents

0=1.

which is impossible.

Therefore

$$\boxed{\text{No Solution}}$$

Summary Table

Case	REF/RREF Appearance	Result
Unique Solution	No. of equations = No. of variables	Exactly one solution
Infinite Solutions	No. of variables is more than No. of equations, i.e. in at least one row (0=0) exist	Infinitely many solutions
No Solution	There is a Row of the form $(0 = b, b \in \mathbb{R}, b \neq 0)$	No solution

Memory Trick

$$\boxed{0 = 0}$$

means **good news** (dependent equations → infinitely many solutions).

$$\boxed{0 = b}$$

means **bad news** (contradiction → no solution).

Everything else typically leads to a **unique solution**.

Exercise

Solve the following system of linear equations using **Gaussian Elimination Method** and **Gauss–Jordan Elimination Method**:

1.

$$2x + y = 5,$$
$$x - y = 1.$$

2.

$$2x + 4y = 8,$$
$$x + 2y = 4.$$

3.

$$2x + 3y = 5,$$
$$4x + 6y = 12.$$

Solve the following system using Gaussian and Gauss–Jordan Elimination:

Problem 1:

$$x + y + z = 6,$$
$$2x + 3y + z = 10,$$
$$x + 2y + 3z = 14.$$

Solution:

Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 10 \\ 1 & 2 & 3 & 14 \end{array}\right]$$

Gaussian Elimination (REF)

Apply:

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -2 \\ 0 & 1 & 2 & 8 \end{array}\right]$$

Next,

$$R_3 \to R_3 - R_2$$

$$\boxed{\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 3 & 10 \end{array}\right]}$$

This is REF.

Back Substitution

From the third row,

$$3z = 10$$

$$z = \frac{10}{3}$$

From the second row,

$$y - z = -2$$

$$y = \frac{4}{3}$$

From the first row,

$$x+y+z=6$$

$$x+\frac{4}{3}+\frac{10}{3}=6$$

$$x=\frac{4}{3}$$

Therefore,

$$\boxed{x=\frac{4}{3},\quad y=\frac{4}{3},\quad z=\frac{10}{3}}$$

Gauss–Jordan Elimination (RREF)

Continue from REF:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 3 & 10 \end{array}\right]$$

Make the third pivot 1:

$$R_3 \to \frac{1}{3}R_3$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & \frac{10}{3} \end{array}\right]$$

Eliminate above:

$$R_2 \rightarrow R_2 + R_3$$

$$R_1 \rightarrow R_1 - R_3$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & \frac{8}{3} \\ 0 & 1 & 0 & \frac{4}{3} \\ 0 & 0 & 1 & \frac{10}{3} \end{array}\right]$$

Finally,

$$R_1 \rightarrow R_1 - R_2$$

$$\boxed{\left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{4}{3} \\ 0 & 1 & 0 & \frac{4}{3} \\ 0 & 0 & 1 & \frac{10}{3} \end{array}\right]}$$

Hence,

$$\boxed{x = \frac{4}{3},\ y = \frac{4}{3},\ z = \frac{10}{3}}$$

Problem 2:

$$x + y + z = 3,$$
$$2x + 2y + 2z = 6,$$
$$x - y + z = 1.$$

Solution:

Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 2 & 2 & 6 \\ 1 & -1 & 1 & 1 \end{array}\right]$$

Gaussian Elimination

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - R_1$$

gives

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & -2 \end{array}\right]$$

Interchange R_2 and R_3:

$$\boxed{\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & -2 & 0 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right]}$$

REF obtained.

Back Substitution

Second row:

$$-2y = -2$$

$$y = 1$$

First row:

$$x + 1 + z = 3$$

$$x + z = 2$$

Let

$$z = t$$

Then,

$$x = 2 - t$$

Therefore,

$$\boxed{x = 2 - t, \quad y = 1, \quad z = t}$$

where $t \in \mathbb{R}$.

Gauss–Jordan Elimination

From REF:

$$R_2 \to -\frac{1}{2}R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Eliminate above the pivot:

$$R_1 \to R_1 - R_2$$

$$\boxed{\left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]}$$

Hence,

$$x + z = 2, \quad y = 1$$

Let $z = t$,

$$\boxed{(x, y, z) = (2 - t, 1, t)}$$

Problem 3:

$$x + y + z = 2,$$
$$2x + 2y + 2z = 5,$$
$$x - y + z = 1.$$

Solution:

Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 2 & 2 & 5 \\ 1 & -1 & 1 & 1 \end{array}\right]$$

Gaussian Elimination

Apply:

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - R_1$$

we get

$$\boxed{\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & -2 & 0 & -1 \end{array}\right]}$$

Interchange R_2 and R_3:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

The last row gives

$$0x + 0y + 0z = 1$$

or

$$\boxed{0 = 1}$$

which is impossible.

Hence,

$$\boxed{\text{The system is inconsistent and has no solution.}}$$

Gauss–Jordan Elimination

The row

$$[0\ 0\ 0 \mid 1]$$

cannot be transformed into a valid equation.

Therefore, the RREF also contains the contradictory row

$$\boxed{0 = 1}$$

and the system has **no solution**.

Problem 4: Solve the system of equations using Gauss–Jordan Elimination Method.

$$x + y + z = 2,$$
$$2x + 2y + 2z = 5,$$
$$x - y + z = 1.$$

Solution:

Step 1: Write the Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 2 & 2 & 5 \\ 1 & -1 & 1 & 1 \end{array}\right]$$

Step 2: Make zeros below the first pivot

Apply the row operations:

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - R_1$$

Therefore,

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & -2 & 0 & -1 \end{array}\right]$$

Step 3: Interchange rows to obtain a proper echelon arrangement

Apply:

$$R_2 \leftrightarrow R_3$$

Hence,

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

Step 4: Make the second pivot equal to 1

Apply:

$$R_2 \to -\frac{1}{2}R_2$$

Thus,

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{array}\right]$$

Step 5: Eliminate the entry above the second pivot

Apply:

$$R_1 \to R_1 - R_2$$

Therefore,

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & \frac{3}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{array}\right]$$

Step 6: Interpret the RREF

The last row represents the equation

$$0x + 0y + 0z = 1,$$

or equivalently,

$$0 = 1.$$

This is a contradiction.

Conclusion

Since the reduced row echelon form contains a contradictory row,

$$\boxed{[0\ 0\ 0 \mid 1]},$$

the system is **inconsistent**. Therefore, the given system of linear equations has

$$\boxed{\text{No solution.}}$$

Exercise Set: Systems of Three Linear Equations

1.

$$\begin{aligned} &x + y + z = 9, \\ &2x + 3y + z = 13, \\ &x + 2y + 3z = 17. \end{aligned}$$

2.

$$\begin{aligned} &x + y + z = 6, \\ &2x + 2y + 2z = 12, \\ &x - y + z = 2. \end{aligned}$$

3.

$$\begin{aligned} &x + y + z = 3, \\ &2x + 2y + 2z = 7, \\ &x - y + z = 1. \end{aligned}$$

Problem 1: A 2×3 System

Solve using Gaussian and Gauss–Jordan Elimination:

$$\begin{aligned} x + y + z &= 6, \\ 2x + 3y + z &= 10. \end{aligned}$$

Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 10 \end{array}\right]$$

Gaussian Elimination

$$R_2 \rightarrow R_2 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -2 \end{array}\right]$$

This is REF.

From the second row,

$$y - z = -2$$

Let

$$z = t$$

Then,

$$y = t - 2$$

From the first row,

$$x + (t - 2) + t = 6$$

Therefore,

$$x = 8 - 2t$$

Hence,

$$\boxed{(x, y, z) = (8 - 2t, t - 2, t), \quad t \in \mathbb{R}}$$

Gauss–Jordan Elimination

From the REF,

$$R_1 \to R_1 - R_2$$

giving

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 8 \\ 0 & 1 & -1 & -2 \end{array}\right]$$

This is RREF.

Thus,

$$x + 2z = 8, \quad y - z = -2$$

Let $z = t$.

Hence,

$$\boxed{(x, y, z) = (8 - 2t, t - 2, t)}$$

Problem 2: A 3×4 System

Solve using Gaussian and Gauss–Jordan Elimination:

$$x + y + z + w = 8$$

$$2x + 3y + z + w = 12$$

$$x + 2y + 3z + w = 14$$

Gaussian Elimination

Step 1: Write the augmented matrix

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 8 \\ 2 & 3 & 1 & 1 & 12 \\ 1 & 2 & 3 & 1 & 14 \end{array}\right]$$

Step 2: Make zeros below the first pivot

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - R_1$$

giving

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 8 \\ 0 & 1 & -1 & -1 & -4 \\ 0 & 1 & 2 & 0 & 6 \end{array}\right]$$

Step 3: Make zero below the second pivot

$$R_3 \to R_3 - R_2$$

Hence,

$$\boxed{\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 8 \\ 0 & 1 & -1 & -1 & -4 \\ 0 & 0 & 3 & 1 & 10 \end{array}\right]}$$

This is the **Row Echelon Form (REF)**.

Step 4: Back Substitution

Since there are four variables and only three pivot equations, one variable is free.

Let

$$w = t, \qquad t \in \mathbb{R}.$$

From the third row,

$$3z + w = 10$$

or,

$$3z + t = 10$$

Therefore,

$$z = \frac{10 - t}{3}.$$

From the second row,

$$y - z - w = -4$$

or,

$$y-\frac{10-t}{3}-t=-4.$$

Therefore,

$$\begin{aligned} y &= \frac{10-t}{3}+t-4 \\ &= \frac{10-t+3t-12}{3} \\ &= \frac{2t-2}{3}. \end{aligned}$$

From the first row,

$$x+y+z+w=8.$$

Therefore,

$$x+\frac{2t-2}{3}+\frac{10-t}{3}+t=8.$$

Hence,

$$x + \frac{2t - 2 + 10 - t}{3} + t = 8$$

$$x + \frac{t + 8}{3} + t = 8$$

$$x + \frac{4t + 8}{3} = 8$$

Thus,

$$x = 8 - \frac{4t + 8}{3}$$

$$= \frac{24 - 4t - 8}{3}$$

$$= \frac{16 - 4t}{3}.$$

Therefore,

$$\boxed{x = \frac{16 - 4t}{3}, \quad y = \frac{2t - 2}{3}, \quad z = \frac{10 - t}{3}, \quad w = t}$$

where

$$t \in \mathbb{R}.$$

Thus, the system has **infinitely many solutions.**

Gauss–Jordan Elimination

Starting from the REF:

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 8 \\ 0 & 1 & -1 & -1 & -4 \\ 0 & 0 & 3 & 1 & 10 \end{array}\right]$$

Step 1: Make the third pivot equal to 1

$$R_3 \to \frac{1}{3}R_3$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 8 \\ 0 & 1 & -1 & -1 & -4 \\ 0 & 0 & 1 & \frac{1}{3} & \frac{10}{3} \end{array}\right]$$

Step 2: Eliminate the entry above the third pivot

$$R_2 \to R_2 + R_3$$

$$R_1 \to R_1 - R_3$$

giving

$$\left[\begin{array}{cccc|c} 1 & 1 & 0 & \frac{2}{3} & \frac{14}{3} \\ 0 & 1 & 0 & -\frac{2}{3} & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{1}{3} & \frac{10}{3} \end{array}\right]$$

Step 3: Eliminate the entry above the second pivot

$$R_1 \to R_1 - R_2$$

Hence, the RREF becomes

$$\boxed{\left[\begin{array}{cccc|c} 1 & 0 & 0 & \frac{4}{3} & \frac{16}{3} \\ 0 & 1 & 0 & -\frac{2}{3} & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{1}{3} & \frac{10}{3} \end{array}\right]}$$

Interpretation of RREF

The equations are

$$x + \frac{4}{3}w = \frac{16}{3},$$

$$y - \frac{2}{3}w = -\frac{2}{3},$$

$$z + \frac{1}{3}w = \frac{10}{3}.$$

Let

$$w = t.$$

Therefore,

$$\boxed{x = \frac{16-4t}{3}, \quad y = \frac{2t-2}{3}, \quad z = \frac{10-t}{3}, \quad w = t}$$

where

$$t \in \mathbb{R}.$$

Problem: A 3×2 System

Solve the following system using **Gaussian Elimination** and **Gauss–Jordan Elimination**:

$$x + y = 5$$

$$2x - y = 1$$

$$3x + 2y = 11$$

Gaussian Elimination Method

Step 1: Write the augmented matrix

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 2 & -1 & 1 \\ 3 & 2 & 11 \end{array}\right]$$

Step 2: Make zeros below the first pivot

Apply the row operations:

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - 3R_1$$

Thus,

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -3 & -9 \\ 0 & -1 & -4 \end{array}\right]$$

Step 3: Make zero below the second pivot

Apply:

$$R_3 \to 3R_3 - R_2$$

Therefore,

$$\boxed{\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -3 & -9 \\ 0 & 0 & -3 \end{array}\right]}$$

Interpretation

The last row represents

$$0x + 0y = -3,$$

or,

$$0 = -3.$$

This is a contradiction.

Conclusion

The system is **inconsistent** and has **no solution**.

Problem: 3×2 System (Unique Solution)

Solve the following system using **Gaussian Elimination** and **Gauss–Jordan Elimination**:

$$x + y = 4$$

$$2x - y = 5$$

$$3x + 4y = 13$$

Gaussian Elimination Method

Step 1: Write the augmented matrix

$$\left[\begin{array}{cc|c} 1 & 1 & 4 \\ 2 & -1 & 5 \\ 3 & 4 & 13 \end{array}\right]$$

Step 2: Eliminate entries below the first pivot

Apply:

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - 3R_1$$

Thus,

$$\left[\begin{array}{cc|c} 1 & 1 & 4 \\ 0 & -3 & -3 \\ 0 & 1 & 1 \end{array}\right]$$

Step 3: Eliminate the entry below the second pivot

Apply:

$$R_3 \to 3R_3 + R_2$$

Therefore,

$$\boxed{\left[\begin{array}{cc|c} 1 & 1 & 4 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{array}\right]}$$

This is the **Row Echelon Form (REF)**.

Step 4: Back Substitution

From the second row:

$$-3y = -3$$

Therefore,

$$y = 1$$

From the first row:

$$x + y = 4$$

Hence,

$$x + 1 = 4$$

Therefore,

$$x = 3$$

Solution

$$\boxed{x = 3, \quad y = 1}$$

Gauss–Jordan Elimination Method

Start from the REF:

$$\left[\begin{array}{cc|c} 1 & 1 & 4 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{array}\right]$$

Step 1: Make the second pivot equal to 1

Apply:

$$R_2 \to -\frac{1}{3}R_2$$

Thus,

$$\left[\begin{array}{cc|c} 1 & 1 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right]$$

Step 2: Eliminate the entry above the second pivot

Apply:

$$R_1 \to R_1 - R_2$$

Therefore,

$$\boxed{\left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right]}$$

This is the **Reduced Row Echelon Form (RREF)**.

Final Answer

$$\boxed{x = 3, \qquad y = 1}$$

Exercise Set: Non-Square Systems of Linear Equations

Instructions:
Solve each of the following systems of linear equations using **both Gaussian Elimination and Gauss–Jordan Elimination**.

1.

$$x + y + z = 7$$

$$2x + 3y + z = 12$$

2.

$$x + y + z + w = 10$$

$$2x + 3y + z + w = 16$$

$$x + 2y + 3z + w = 20$$

3.

$$x + 2y = 8$$

$$2x - y = 4$$

$$3x + 3y = 12$$

4.

$$x + y + z = 6$$

$$2x - y + z = 3$$

$$x + 3y + 2z = 11$$

$$3x + 2y + 3z = 14$$

5.

$$x + y = 5$$

$$2x - y = 1$$

$$3x + 2y = 12$$

6.

$$x + y + z = 4$$

$$2x + 3y - z = 5$$

$$x - y + 2z = 7$$

$$3x + 2y + z = 20$$

Homogeneous System of Linear Equations

A system of linear equations is called **homogeneous** if all constant terms are zero. In matrix form,

$$AX = 0$$

where A is the coefficient matrix and 0 is the zero vector.

Important Facts

1. Every homogeneous system has at least one solution:

$$\boxed{X = 0}$$

called the **trivial solution**.

2. A homogeneous system has **non-trivial solutions** if the number of free variables is at least one, i.e.,

$$\text{rank}(A) < n,$$

where n is the number of unknowns.

3. If

$$\text{rank}(A) = n,$$

then the only solution is the trivial solution.

Problem 1: A 2×2 Homogeneous System (Only Trivial Solution)

Solve using Gaussian and Gauss–Jordan Elimination:

$$x + y = 0$$

$$2x - y = 0$$

Gaussian Elimination

Augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 2 & -1 & 0 \end{array}\right]$$

Apply:

$$R_2 \to R_2 - 2R_1$$

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & -3 & 0 \end{array}\right]$$

This is REF.

Back substitution:

$$-3y = 0$$

$$y = 0$$

From the first equation:

$$x + y = 0$$

therefore,

$$x = 0$$

Hence,

$$\boxed{x = 0,\ y = 0}$$

Gauss–Jordan Elimination

From REF:

$$R_2 \to -\frac{1}{3}R_2$$

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 1 & 0 \end{array}\right]$$

Now,

$$R_1 \to R_1 - R_2$$

giving:

$$\boxed{\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right]}$$

Thus,

$$\boxed{(x, y) = (0, 0)}$$

Problem 2: A 3×3 Homogeneous System (Non-trivial Solution)

Solve:

$$x + y + z = 0$$

$$2x + 2y + 2z = 0$$

$$x - y + z = 0$$

Gaussian Elimination

Augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 \\ 1 & -1 & 1 & 0 \end{array}\right]$$

Apply:

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - R_1$$

giving:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \end{array}\right]$$

Interchange:

$$R_2 \leftrightarrow R_3$$

REF:

$$\boxed{\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]}$$

Back substitution:

$$-2y = 0$$

so,

$$y = 0$$

Let

$$z = t$$

From first equation:

$$x + z = 0$$

therefore,

$$x = -t$$

Hence,

$$\boxed{(x, y, z) = (-t, 0, t), \quad t \in \mathbb{R}}$$

Gauss–Jordan Elimination

From REF:

$$R_2 \to -\frac{1}{2}R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Now,

$$R_1 \to R_1 - R_2$$

Therefore,

$$\boxed{\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]}$$

Thus,

$$x + z = 0, \quad y = 0$$

Let $z = t$,

$$\boxed{(x, y, z) = (-t, 0, t)}$$

Problem 3: A 4×4 Homogeneous System (Non-trivial Solution)

Solve:

$$x + y + z + w = 0$$

$$2x + 2y + 2z + 2w = 0$$

$$x - y + z - w = 0$$

$$3x + y + 3z + w = 0$$

Gaussian Elimination

The augmented matrix is

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 2 & 0 \\ 1 & -1 & 1 & -1 & 0 \\ 3 & 1 & 3 & 1 & 0 \end{array}\right]$$

Apply:

$$R_2 - 2R_1, \quad R_3 - R_1, \quad R_4 - 3R_1$$

we obtain:

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & -2 & 0 \\ 0 & -2 & 0 & -2 & 0 \end{array}\right]$$

After simplification:

$$\boxed{\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & -2 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]}$$

Let:

$$z = s, \quad w = t$$

From the second row:

$$-2y - 2w = 0$$

$$y = -t$$

From the first row:

$$x + y + z + w = 0$$

so,

$$x + s = 0$$

Therefore,

$$x = -s$$

Hence,

$$\boxed{(x, y, z, w) = (-s, -t, s, t)}$$

Problem 4: Non-square Homogeneous System (2×3)

Solve:

$$x + y + z = 0$$

$$2x + 3y + z = 0$$

Gaussian Elimination

Augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & 3 & 1 & 0 \end{array}\right]$$

Apply:

$$R_2 \to R_2 - 2R_1$$

REF:

$$\boxed{\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right]}$$

Let:

$$z = t$$

Then:

$$y = t$$

and

$$x + 2t = 0$$

so,

$$x = -2t$$

Hence,

$$\boxed{(x, y, z) = (-2t, t, t)}$$